

In-silico surface-antigen prioritisation for extraskeletal myxoid chondrosarcoma: one cell line, a translocation-sarcoma surrogate, a hard normal-tissue filter — and a validation request

Preprint status (2026-07-03). Computational, **hypothesis-generating** manuscript, extensively self-red-teamed (see [emc-surface-target-redteam.md](#)). It reports what an honest in-silico surface-antigen analysis for EMC can and cannot establish from public data. The central, deliberately un-triumphant finding is that **a rigorous selectivity test plus a hard normal-tissue-window filter leaves essentially no classic protein surface antigen that is both tumour-selective and normal-tissue-restricted** — the analysis mainly *refines priorities and flags liabilities* (notably that the field-default B7-H3 is not selective in the data), surfaces one real EMC cell line's own profile, and nominates a neuroendocrine-differentiation hypothesis (SSTR2 / GD2) the field has not pursued for EMC. No antigen is asserted as an EMC-validated target. The manuscript's second purpose is a specific, respectful request to the groups holding patient-derived EMC models for the data that would actually resolve it (§7).

Keywords: extraskeletal myxoid chondrosarcoma; EWSR1::NR4A3; surfaceome; therapeutic window; antibody-drug conjugate; CAR-T; radioligand therapy; SSTR2; neuroendocrine differentiation.

Abstract

Background. Extraskeletal myxoid chondrosarcoma (EMC) is a rare translocation sarcoma driven by the *EWSR1::NR4A3* fusion, a nuclear transcription factor. Driver-directed routes (an NR4A3 degrader; a fusion-junction antisense oligonucleotide) confront an intracellular-delivery or druggability gate. Cell-surface antigens offer an orthogonal axis (ADC, CAR/NK, T-cell engager, radioligand therapy) whose gating differs — but at the cost of the fusion-level selectivity the oligonucleotide uniquely offers.

Methods. We built a public-data pipeline: a largely-unbiased human surfaceome (UniProt plasma-membrane + transmembrane/GPI, unioned with a small actionable-antigen seed) ranked by expression and by a rank-based selectivity test (Mann-Whitney, Benjamini-Hochberg) across a translocation-sarcoma DepMap class (n = 76), which — contrary to the common assumption — **also contains one genuine EMC line (H-EMC-SS / ACH-001519)** whose surface transcriptome we report directly (n = 1, descriptive). We then applied a hard **normal-tissue therapeutic-window filter** (Human Protein

Atlas tissue *and* blood-cell specificity, with vital-tissue/immune overrides; controls validated). Positive/negative/hard controls check each classifier branch.

Results. Selectivity (versus other cancer lineages) is significant for CDH11, KIT, FGFR1, NCAM1, GPC2, PTK7, MCAM and EPHB4, and — importantly — **not** for B7-H3/CD276 (BH $q = 1.0$), EGFR or FAP. The normal-tissue filter is decisive and sobering: of the classic candidates, **none is both selective and normal-tissue-restricted**. They partition into selective-but-broadly-expressed (CDH11, GPC2, FGFR1, MCAM, EPHB4), selective-but-immune/vital-liability (NCAM1/CD56 — on NK cells; PTK7; KIT), and non-selective (B7-H3, EGFR, FAP); only GD2 (assayed via its synthase B4GALNT1, an imperfect proxy) shows a restricted *normal-tissue* window (its EMC expression is unmeasured). EMC's reported neuroendocrine differentiation (IHC: INSM1, synaptophysin — pending citation) motivates two hypotheses the field has not tested for EMC: **SSTR2** (target of the approved radioligand ^{177}Lu -DOTATATE) and **GD2** — both requiring direct EMC measurement.

Conclusion. Done rigorously, in-silico surface-target discovery for EMC does not hand over a clean target; its value is to de-risk over-optimistic assumptions (especially B7-H3), to expose antigen-specific liabilities, to surface the one available EMC line's profile, and to nominate the neuroendocrine SSTR2/GD2 route. The decisive validation — EMC surface *protein* expression with a normal-tissue window — requires the patient-derived EMC lines, and we invite their holders to co-resolve it.

1. Introduction

EMC is a rare soft-tissue sarcoma (commonly cited as well under 1% of soft-tissue sarcomas [citation to verify]) defined by a translocation fusing the 5' region of *EWSR1* (less often TAF15/others) to the orphan nuclear receptor *NR4A3*, producing a chimeric transcription factor on a genome with few recurrent secondary mutations [Sjögren; Panagopoulos; whole-genome characterisation citation to verify]. Its line of differentiation was long debated; recent immunohistochemistry supports a **neuroendocrine phenotype** — INSM1 is a sensitive/relatively specific EMC marker and synaptophysin/NSE are frequently expressed, while S100 is only focally positive [Yoshida/Modern Pathology 2017; comprehensive review 2025 — citations to verify].

The intracellular-target bottleneck and the surface alternative. Because the driver is nuclear, driver-directed therapies act inside the cell and confront a druggability or oligonucleotide-delivery gate. Cell-surface antigens enable modalities that change the problem: ADC (clinically validated antibody-mediated delivery), T-cell engagers and CAR-T/NK (act at the surface; no intracellular delivery), and radioligand therapy (payload is radiation). We stress at the outset that this is a *different* gating, **not** an obviously easier one in EMC (§5), and that it sacrifices the fusion-exclusivity of the RNA route.

The gap. EMC surface-antigen expression has not been systematically mapped, and EMC has been assumed absent from public cell-line panels. We provide the first in-silico surface-antigen

analysis for EMC, with a deliberately hard normal-tissue filter and honest statistics, to prioritise scarce validation and to specify what that validation is.

2. Methods

All analyses are computational, use public data, run in CI, and commit their outputs (Data & code). No wet-lab work was done. Provenance and per-stage controls are recorded in each output JSON.

2.1 Surfaceome. UniProt-reviewed human proteins with a plasma-membrane location (SL-0039) plus a transmembrane (KW-0812) or GPI-anchor (KW-0336) topology, unioned with a curated seed of actionable surface antigens (so established targets are always evaluated). The committed run used **2,820 genes from UniProt $\cup$ 47 curated seed antigens** (41 of the seed already in the UniProt set) = **2,826 unique**, of which **2,692** are present in the DepMap expression matrix and were scanned — so the seed is a small, mostly-redundant minority and the set is largely, though **not strictly, unbiased**; the UniProt fetch status and counts are recorded in the output. [emc_surfaceome_scan.py]

2.2 Expression, selectivity, and the one EMC line. DepMap OmicsExpression ($\log_2(\text{TPM}+1)$). We defined a translocation-sarcoma class by OncotreeSubtype (Ewing/synovial/alveolar/DSRCT/clear-cell + the EMC line; $n = 76$) as a lineage-generic surrogate, and — a correction to the field’s assumption — identified a **single genuine EMC line, ACH-001519 (H-EMC-SS), OncotreeSubtype “Extraskelatal Myxoid Chondrosarcoma”**, whose own top surface antigens we report ($n = 1$, descriptive; H-EMC-SS authentication and EWSR1::NR4A3 status flagged [to verify]). For each surface gene we report expression, an effect size (enrichment versus non-sarcoma lineages), and a **rank-based one-sided Mann-Whitney p that the class exceeds the rest, Benjamini-Hochberg corrected**. We are explicit that this is cross-*cancer* selectivity (a distinguishable-from-epithelial-tumours descriptor), **not** a tumour-vs-normal window, and that it mechanically favours mesenchymal antigens because the DepMap panel is epithelial-dominated. Self-check: housekeeping genes are excluded by construction (a minimal sanity check, not validation); CD276 recovers as broadly expressed. [emc_surfaceome_scan.py]

2.3 Normal-tissue therapeutic-window filter (the primary axis). For each antigen we queried the Human Protein Atlas for RNA tissue specificity, tissue distribution, per-tissue nTPM, **blood-cell specificity**, and subcellular location, and classified a window verdict with correct HPA semantics: only *tissue enriched / group enriched* with a *restricted distribution* and no vital-tissue and no strong immune/circulating signal is **RESTRICTED**; *tissue enhanced* (detected broadly with a peak) is **ENHANCED_BROAD**, not restricted; *low tissue specificity* or a **“detected in all” distribution** is **BROAD LIABILITY** (the distribution override demotes e.g. MCAM, which is “group enriched” but detected in all tissues); and expression in a vital tissue, or a *confined blood signal (immune-cell enriched / group enriched* — not weak *immune-cell enhanced*), overrides all as **VITAL OR IMMUNE LIABILITY**. Controls: DLL3/GPC3 (tumour-restricted) must be RESTRICTED, B2M BROAD, and a **hard control**

CD3E (an immune antigen) must NOT be RESTRICTED — all satisfied. HPA RNA is bulk normal tissue and a window *prior*, not a safety guarantee, and mRNA $\neq$ surface protein. [emc_surface_normal_window.py]

2.4 Real-EMC public data (attempted). We attempted the only usable dedicated public EMC tumour transcriptome we could identify (GSE4303) and report why it is unusable. [emc_gse4303_crosscheck.py, emc_line_data_probe.py]

3. Results

3.1 The one EMC line in public data — H-EMC-SS

Contrary to the assumption that EMC is absent from public cell-line panels, DepMap contains one line annotated Extraskelatal Myxoid Chondrosarcoma — **ACH-001519 / H-EMC-SS**. Its own top surface-antigen transcripts are the most EMC-specific in-silico signal available ($n = 1$, descriptive; no statistics; authentication/fusion status [to verify]):

Table 1. H-EMC-SS (ACH-001519) top surface-antigen transcripts (log2(TPM+1); single sample, descriptive; [emc-surfaceome-scan.json $\rightarrow$ emc_line_top_surface]).

Rank	Antigen (log2TPM)	Rank	Antigen (log2TPM)
1	APP (9.9)	9	CD164 (8.5)
2	CD63 (9.5)	10	DNER (8.5) <i>neural</i>
3	FGFR1 (9.3)	11	BSG/CD147 (8.2)
4	SLC38A2 (9.0)	12	RTN4/Nogo (8.2) <i>neural</i>
5	GPRC5B (8.9)	13	MMP14 (8.1)
6	PERP (8.8)	14	ITGB1 (7.9)
7	SLC3A2 (8.6)	15	PMP22 (7.8) <i>neural/myelin</i>
8	CD81 (8.5)	16	ALCAM (7.7)

Two honest readings. First, the list is **dominated by ubiquitous membrane proteins** (APP, the tetraspanins CD63/CD81, BSG/CD147, ITGB1, ALCAM) — raw single-line expression surfaces housekeeping surface proteins, which is precisely why the selectivity and normal-tissue filters below are necessary and why “highly expressed” is not “targetable”. Second, **FGFR1 ranks third** and several **neural-associated surface proteins (DNER, RTN4/Nogo, PMP22)** appear — loosely consistent with EMC’s neuroendocrine/neural differentiation (§1), though from a single line this is a suggestion, not evidence, for the SSTR2/GD2 hypothesis (§3.4). This one line cannot carry statistical weight; it anchors the analysis in an established EMC line (ECACC-catalogued H-EMC-SS; authentication/fusion status [to verify]) and is exactly the data that, at $n \gg 1$, would resolve the questions below.

3.2 Selectivity across the translocation-sarcoma class — and B7-H3 is not selective

With a rank-based, BH-corrected test (surrogate class vs non-sarcoma lineages; [emc-surfaceome-scan.json]):

Antigen	Enrichment (log2TPM, class – rest)	BH q	Significantly selective?
KIT	+2.46	~0	yes
CDH11	+3.18	~0	yes
FGFR1	+1.99	~0	yes
NCAM1/CD56	+1.74	~0	yes
GPC2	+1.49	~0	yes
PTK7	+1.24	2e-4	yes
MCAM/CD146	+1.09	3e-3	yes
EPHB4	+1.0	3e-4	yes
B7-H3/CD276	+0.14	1.0	NO
FAP	+0.02	0.16	no
EGFR	–2.21	1.0	no

Two cautions, stated up front. First, this is **cross-cancer** selectivity, which the epithelial-dominated DepMap panel biases toward mesenchymal antigens (CDH11's +3.18 is largely “not expressed by carcinomas”, not “tumour-specific”). Second, mRNA magnitude to two decimals conveys false precision about surface-protein density; read these as **coarse tiers**. (Note also that the single EMC line of §3.1 is one of the 76 class lines, so its ~1/76 contribution is negligible and FGFR1 appearing in both §3.1 and §3.2 is *not* independent corroboration.) The one decision-grade result here is negative and useful: **B7-H3, the field's default EMC surface target, is not significantly selective in the data** (q = 1.0). B7-H3 protein can be tumour-restricted despite broad mRNA — the basis of its clinical traction — so this is a selectivity-of-transcript caveat, not a claim about protein; but it removes the transcriptomic rationale for treating B7-H3 as the obvious first choice.

3.3 The normal-tissue window is decisive — and leaves no clean classic antigen

Applying the hard window filter (controls validated: DLL3/GPC3 → RESTRICTED, B2M → BROAD, CD3E → VITAL_OR_IMMUNE_LIABILITY; [emc-surface-normal-window.json]):

Antigen	HPA tissue specificity	Blood-cell signal	Wir
B4GALNT1 (GD2 synthase)	Tissue enriched	none	RESTRICTED
CDH11	Tissue enhanced	none	ENHANCED
GPC2	Tissue enhanced	none	ENHANCED
FAP	Tissue enhanced	none	ENHANCED
	Tissue		

SSTR2	enhanced	none	ENHANCED
EGFR	Tissue enhanced	none	ENHANCED
FGFR1	Low tissue specificity	—	BROAD_LIAI
MCAM/CD146	Group enriched	none	BROAD_LIAI
EPHB4	Low tissue specificity	immune enhanced	BROAD_LIAI
CD276/B7-H3	Low tissue specificity	none	BROAD_LIAI
ERBB2	Low tissue specificity	immune enhanced	BROAD_LIAI
NCAM1/CD56	Tissue enhanced	immune enriched (NK)	VITAL_OR_I
PTK7	Low tissue specificity	immune enriched	VITAL_OR_I
KIT	Tissue enhanced	group enriched (haematopoietic/mast)	VITAL_OR_I

The intersection of §3.2 (selective) and §3.3 (restricted) among classic protein antigens is **empty** (**Figure 1**: antigens plotted by selectivity × window tier; the selective-and-restricted quadrant is unpopulated — [emc-surface-prioritization.png]). The candidates fail the window in specific, nameable ways: - **NCAM1/CD56** is on NK cells (fratricide risk for CAR/NK; a circulating compartment) and neural tissue; the CD56 ADC **lorvotuzumab mertansine (IMGN901)** was clinically developed and discontinued (no efficacy benefit, added toxicity) [Socinski/Spira — citations to verify]. It is not a clean target despite its selectivity. - **CDH11** is broadly expressed in normal fibroblasts, synovium and bone (an established rheumatoid-arthritis/ fibrosis target); its high cross-cancer enrichment is the mesenchymal-vs-epithelial artifact above. Pairing it with binary-kill modalities (CAR/TCE) would attack normal mesenchyme body-wide. - **B7-H3, EGFR, FAP** are non-selective and/or broad; **FGFR1/MCAM/EPHB4** are window-liabilities. - **GD2** (via the B4GALNT1 proxy — GD2 is a glycolipid, not a gene product, so this is indirect) has the only restricted *normal-tissue* window here, consistent with GD2's known tumour-restricted profile — but **whether EMC expresses GD2 is unmeasured**, so this is a favourable window, not evidence of an EMC target.

3.4 A neuroendocrine hypothesis the field has not tested for EMC: SSTR2 and GD2

EMC's neuroendocrine differentiation (INSM1+, synaptophysin+; §1) motivates two candidate targets absent from prior EMC surface discussions: - **SSTR2 (somatostatin receptor 2)** — the target of the *approved* radioligand ¹⁷⁷**Lu-DOTATATE** (Lutathera; approved for gastroenteropancreatic neuroendocrine tumours on the NETTER-1 trial [Strosberg et al., *N Engl J Med* 2017 — verify]) and its ²²⁵Ac α-analogues. If EMC's neuroendocrine phenotype extends to SSTR2 surface expression, an off-the-shelf theranostic (SSTR-PET + peptide-receptor radioligand therapy) becomes testable without a bespoke

agent. The first gate is whether EMC expresses SSTR2 at all (unmeasured); *if* it does, then — its window being ENHANCED_BROAD (SSTR2 is expressed in normal neuroendocrine/GI tissue) — dosimetry, not novelty, is the remaining gate, as for approved NET-RLT. - **GD2** — a surface glycolipid with mature CAR/antibody platforms and (via B4GALNT1) the only restricted-window signal in §3.3.

These are hypotheses requiring direct EMC measurement (SSTR2 IHC/PET; GD2 immunostaining), not claims.

3.5 The public real-EMC tumour dataset is unusable

The only usable, dedicated public EMC tumour transcriptome we could identify, GSE4303, is a seven-platform two-colour cDNA-clone microarray (three EMC samples per platform; `matrix_files_found` lists seven GPLs) whose values are reference-pool log-ratios and whose probes lack gene symbols; zero shortlist genes resolved (`[emc-gse4303-crosscheck.json]`). It cannot rank surface antigens. Combined with §3.1 (a single EMC line in DepMap), this is precisely why patient-derived EMC model data is the essential input (§7).

4. The modality axis — differently gated, not obviously easier

If a validated EMC surface antigen existed, the modalities it enables gate differently from the oligonucleotide:

Modality	Intracellular delivery?	Replacing gate	Crossfire note
T-cell engager (antigen x CD3)	No	Cold/excluded TME (poor in adult sarcoma)	—
CAR-T / CAR-NK	No	Solid-tumour infiltration; cold TME; fratricide (for NK-cell antigens)	—
Radioligand therapy	No	Antigen level; dosimetry; tumour-to-normal uptake ratio	β (^{177}Lu) ~2 mm; α (^{225}Ac) ~50-80 μm
ADC	Internalisation (clinically validated)	Internalisation; tumour-vs-normal window	—

Correcting a common intuition: radioligand crossfire mitigates *heterogeneous tumour uptake* (it kills tumour cells the ligand never bound); it does **not** make a *broadly-expressed normal* antigen safer — for a normal-tissue antigen, crossfire widens the irradiated field. So B7-H3’s breadth is not “rescued” by RLT; the RLT case rests on tumour-to-normal uptake ratios and dosimetry (the basis of approved SSTR2-RLT).

5. Discussion

A differently-gated third axis, honestly weighed. EMC's driver can be attacked from protein (degrader), RNA (fusion-exclusive ASO) and surface (this work) compartments. The surface axis trades the ASO's intracellular-delivery gate for others that, in EMC specifically, may be no easier: the abundant **myxoid/ chondroid extracellular matrix** is a diffusion and binding-site barrier to antibodies, adoptive cells and radioligands, and adult sarcoma has a poor CAR/TCE record in cold, immune-excluded tumours. The honest framing is *orthogonal and differently-gated*, not "more tractable".

No fusion linkage. Every antigen here is a generic lineage antigen with no mechanistic connection to EWSR1::NR4A3; success or failure would be independent of the fusion. This is a first-order cost: the surface axis abandons the program's distinctive fusion-exclusivity. Its justification is pragmatic (delivery), not mechanistic, and EMC's role is as a clean single-driver *entry* indication for antigens that are pan-sarcoma/ pan-cancer.

What the analysis is good for. Negative and prioritising results: B7-H3 is not transcriptionally selective; CD56/CDH11/PTK7/KIT carry specific window liabilities; the neuroendocrine phenotype points to SSTR2/GD2. These sharpen where scarce wet-lab effort should and should not go.

6. Limitations

- **One EMC line, n = 1** (H-EMC-SS); no statistics from it, and its authentication/fusion status is unverified.
 - **Surrogate is lineage-generic;** surface phenotype tracks lineage, and Ewing/synovial differ from EMC.
 - **mRNA ≠ surface protein;** target density, epitope accessibility and internalisation — what modalities need — are not measured. Rankings are coarse tiers.
 - **Cross-cancer selectivity ≠ tumour-vs-normal;** the normal-tissue window (HPA bulk RNA) is a prior, not a safety guarantee, and GD2 is assessed only via a synthase proxy.
 - **The public EMC dataset is unusable** (§3.5); the surrogate cannot be corroborated against real EMC tumours.
 - **Clinical-agent and EMC-biology citations are flagged** [citation to verify] and must be sourced before use.
 - **Delivery is not solved by naming an antigen;** modality-specific gates (myxoid-matrix penetration, cold TME, dosimetry) remain.
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7. The validation this needs — a collaboration request

This analysis produces *priors and cautions*, not a validated target, and the single decisive dataset is EMC surface **protein** expression with a normal-tissue comparison. Two groups hold patient-derived EMC models: **USZ-EMC** (Bangerter et al., *Human Cell* 2022/2023) and

NCC-EMC1-C1 (Iwata et al., *Human Cell* 2025); DepMap additionally holds H-EMC-SS (n = 1). We propose a genuine collaboration rather than a data extraction:

- If useful to them, we will **run their existing RNA-seq/expression** through this pipeline and share everything; at $n \gg 1$ it validates or refutes the selectivity and window priors on real EMC.
- A targeted **surface panel** — prioritising the questions this analysis raises: is **SSTR2** expressed ($\rightarrow$ off-the-shelf DOTATATE theranostic)? is **GD2** present? and, given the window liabilities, are the selective-but-broad antigens (GPC2, CDH11) actually surface-restricted at the protein level in EMC?
- Authorship and study design led by the groups whose models make the work possible.

We approach the model-holders directly and privately; this manuscript states the analysis, not a claim on their data.

8. Conclusion

A deliberately hard in-silico analysis does not deliver a clean EMC surface target: rigorous selectivity testing plus a normal-tissue window shows the field-default B7-H3 is not selective and that the selective candidates carry specific window liabilities, leaving a favourable-normal-tissue-window GD2 (EMC expression unknown) and a grounded-but-unmeasured-in-EMC SSTR2/DOTATATE neuroendocrine hypothesis as the questions most worth testing. We report the one available EMC line's profile, specify the protein-level validation that matters, and invite the EMC-model groups to resolve it.

Data & code availability

Code and committed outputs (research/modalities/, refreshed to the modalities-cache branch): - Surfaceome scan + selectivity + EMC-line profile — `emc_surfaceome_scan.py` $\rightarrow$ `emc-surfaceome-scan.json` - Normal-tissue window — `emc_surface_normal_window.py` $\rightarrow$ `emc-surface-normal-window.json` - EMC-line data probe — `emc_line_data_probe.py` $\rightarrow$ `emc-line-data-probe.json` - GSE4303 cross-check — `emc_gse4303_crosscheck.py` $\rightarrow$ `emc-gse4303-crosscheck.json` - Red-team log — `emc-surface-target-redteam.md`

Sources: UniProt; DepMap (incl. H-EMC-SS/ACH-001519); Human Protein Atlas; NCBI GEO (GSE4303).

References

Verified in the repository pool: - **Sjögren H, et al.** EWSR1/NR4A3 fusion in EMC. - **Panagopoulos I, et al.** EMC fusion variants/partners. - **Bangerter, et al.** USZ-EMC patient-derived models. *Human Cell* 2022/2023. - **Iwata S, et al.** NCC-EMC1-C1. *Human Cell* 2025. - **Uhlén M, et al.** Human Protein Atlas. *Science* 2015. - **Bausch-Fluck D, et al.** The in silico human surfaceome. *PNAS* 2018.

To verify (candidate anchors identified; confirm before treating as established): - EMC neuroendocrine differentiation — **INSM1** as a sensitive/relatively specific EMC marker (*Modern Pathology* 2017; EMC INSM1 series PMID 36563884) and frequent synaptophysin/NSE positivity; S100 only focal — **[verify]**. - H-EMC-SS (ACH-001519) — an ECACC-catalogued EMC line; authentication STR / EWSR1::NR4A3 status — **[verify via Cellosaurus/DepMap]**. - CD56 ADC **lorvotuzumab mertansine (IMGN901)** — SCLC Phase 1/2 (no efficacy benefit, added toxicity; PMID 28341109) and CD56⁺ solid-tumour Phase I (PMID 26961907) — **[verify]**. - SSTR2 / ¹⁷⁷Lu-**DOTATATE** (Lutathera; NETTER-1, **Strosberg et al., NEJM 2017**) and ²²⁵Ac somatostatin-receptor α -analogues — **[verify]**. - α (²²⁵Ac ~50–80 μ m) vs β (¹⁷⁷Lu ~2 mm) tissue ranges — standard radiobiology — **[verify]**. - CDH11 normal fibroblast/synovial/bone expression (rheumatoid-arthritis/fibrosis target) — **[verify]**. - Clinical-stage agents per antigen (GPC2, PTK7, B7-H3 ifinatamab deruxtecan, etc.) — **[verify]** per antigen. - EMC incidence (<1% of soft-tissue sarcoma) — **[verify]**.

Provenance: consolidates the surfaceome scan (incl. H-EMC-SS profile + BH-corrected selectivity), the normal-tissue window (controls validated), the EMC-line data probe and the GSE4303 cross-check (committed CPU outputs on modalities-cache), and two red-team passes ([emc-surface-target-redteam.md]). No antigen is asserted as an EMC-validated target.